

Changrun Jia

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EDUCATION

Master of Information Technology in Artificial Intelligence

University of New South Wales (UNSW), Sydney, Australia, 2026

- Distinction level at UNSW, denotes superior academic performance.

Relevant Modules:

Artificial Intelligence, Machine Learning & Data Mining, Deep Learning, Reinforcement Learning, Computer Vision, Algorithm Design, Knowledge Representation, Human Computer Interaction, Eng PGCW Research Skills.

Bachelor of Engineering in Software Engineering

Guangzhou College of Commerce, Guangzhou, China, 2021

RESEARCH EXPERIENCE

Principal Investigator (2020–2022)

Guangdong Provincial Science and Technology Innovation Strategic Special Fund, Hippocampus MRI Segmentation

- Awarded RMB 15,000 in provincial research funding to lead a two-year project on deep-learning-based hippocampus MRI segmentation.
- Designed and implemented a medical image segmentation pipeline for hippocampal MRI volumes, including data preprocessing, model development, training, evaluation, and result analysis.
- Investigated 3D CNNs, U-Net-based architectures, and residual learning methods for anatomical structure segmentation in brain MRI.
- Evaluated segmentation performance using medical image analysis metrics and prepared research reports on model behaviour, experimental results, and limitations.

Team Member (2018–2019)

3D U-Net-Based Detection and Segmentation of Human Hippocampus MRI Volumes

- Contributed to a medical image segmentation project focused on detecting and segmenting human hippocampus structures from volumetric MRI data.
- Implemented project modules for 3D MRI preprocessing, segmentation model testing, and experimental result analysis.
- Worked with 3D U-Net-based segmentation methods and evaluation procedures for anatomical structure segmentation.
- Extended this work into subsequent funded research on deep-learning-based hippocampus MRI segmentation.

Team Member (2025)

Aerial Scene Classification Based on KANs Networks

- Designed and implemented comparative experiments for aerial scene classification using KANs and more than ten hybrid deep learning models.
- Built the full experimental pipeline, including image preprocessing, feature modelling, model training, evaluation, and comparative analysis.
- Explored neural architecture design and model comparison for visual representation learning in remote sensing image classification.
- Developed a KAN-based hybrid model that achieved a peak classification accuracy of 97.67

Research Assistant (Part-time, 2019–2021)

Artificial Intelligence and Image Recognition Laboratory, Guangzhou College of Commerce

- Worked as a part-time research assistant in a newly established university laboratory focused on artificial intelligence and image recognition.
- Supported computer vision and image processing research activities, including model implementation, experiment support, and result analysis.
- Assisted with research tasks related to AI-based visual recognition, including literature review, data processing, and experimental documentation.

Project Leader (2019–2021)

Provincial Undergraduate Innovation and Entrepreneurship Project: AI-Based Gait Recognition

- Led a computer vision project on AI-based gait recognition, covering data collection, annotation protocol design, model training, evaluation, and comparative experiments.
- Built an end-to-end deep learning pipeline for human gait recognition from raw data preparation to model validation.
- Coordinated project tasks and experimental planning for model training, evaluation, and comparative analysis.

PUBLICATIONS

First author & corresponding author	Jia, Changrun. A Non-Invasive EEG--EMG--Vision Shared-Control Framework for Reducing Everyday Burden in Upper-Limb Prosthesis Use. <i>2026 IEEE 5th International Conference on Information Technology, Big Data and Artificial Intelligence (ICIBA 2026.5)</i> . Accepted; conference proceedings pending publication.
First author	(In progress) Jia, Changrun. Behaviour-Conditioned Energy-Based Reconstruction of Natural Movies from V1 Population Activity: A Unified DNEM--EBT Framework.
Model design, experiments, manuscript, correspond	Jia, C., Jia, Changrun. , Yu, H. Combining Residual Learning and U-Net for Hippocampus Segmentation of Brain MRI Volume Image. <i>The 2020 4th International Conference on Deep Learning Technologies, 2020, p.94-98; New York, NY, USA: ACM.</i> DOI: 10.1145/3417188.3417191
Design network architecture, training, ablation experiments	Jia, C., Jia, Changrun. , Wei, J. Combining Sequence Learning and Dilated Convolution Network for Hippocampus Segmentation. <i>IEEE Advanced Information Technology, Electronic and Automation Control Conference (IAEAC), 2021, Vol.5, p.844-849.</i> DOI: 10.1109/IAEAC50856.2021.9390902
Algorithm implementation, experimental validation	Jia, C., Jia, Changrun. , Kong, F. Image Saliency Detection Based on Background Prior and Multi-feature Fusion. <i>2020 5th IEEE International Conference on Big Data Analytics (ICBDA), 2020, p.276-281; Xiamen, FJ, CN.</i> DOI: 10.1109/ICBDA49040.2020.9101263
Initial network design, data preprocessing, testing	Jia, C., Liu, D., Kong, F., Jia, Changrun. Three-dimensional Segmentation of Hippocampus in Brain MRI Images Based on 3CN-Net. <i>The 2019 3rd International Conference on Innovation in Artificial Intelligence, 2019, p.17-20; New York, NY, USA: ACM.</i> DOI: 10.1145/3319921.3319941

SELECTED PROJECTS

Deployable Safeguard Multi-Asset AI Financial Research Assistant (Team Member, 2025.9–2026.1)

- Designed and implemented a modular system based on Docker Compose, React, FastAPI, and TimescaleDB.
- Used APScheduler for market, news, and social media data collection and ETL scheduling.
- Integrated FinBERT / Twitter-RoBERTa sentiment analysis and Prophet + XGBoost forecasting services, with configurable support for LSTM / GRU / Transformer.
- Implemented 30-day rolling Pearson correlation analysis and time-series alignment, and completed user-driven evaluation and delivery.

Emotion Recognition and Fine-Grained Classification from Tweets (Team Member, 2025)

- Designed and implemented a Transformer + BiLSTM hybrid model for emotion recognition and fine-grained classification from short, noisy, and informal tweet text.
- Integrated embedding layers, sinusoidal positional encoding, Transformer blocks, BiLSTM layers, residual connections, LayerNorm, and multi-scale pooling strategies.
- Used average pooling, max pooling, and attention pooling to capture both global semantic features and sequential dependencies.
- Achieved a balanced accuracy of 94.95% in tweet-based emotion classification experiments.

National Undergraduate Innovation and Entrepreneurship Training Program: Web Based Face Recognition Attendance System (Project Leader, 2019–2021)

- Developed a web-based face recognition attendance system for university library scenarios.

- Responsible for requirement analysis, algorithm selection and integration, system deployment, debugging, and final project completion.
- Integrated computer vision models into a usable web application, with attention to system reliability, testing, and real-world deployment constraints.

ADDITIONAL EXPERIENCE

CSE Maker Space, Guangzhou College of Commerce (Core Member, 2018–2021)

- Participated in a university-supported innovation laboratory with dedicated space and seed funding.
- Assisted in project application, implementation, and deployment across computer science and electronic information projects.
- Gained extensive experience in AI system prototyping, project collaboration, and interdisciplinary communication.

GRANTS AND FUNDING

- Guangdong Provincial Science and Technology Innovation Strategic Special Fund ("Climbing Plan"), funded with **RMB 15,000**, 2022.
- National Undergraduate Innovation and Entrepreneurship Training Program}, funded with **RMB 10,000**, 2021.
- Provincial Undergraduate Innovation and Entrepreneurship Training Program} AI-Based Gait Recognition, funded with **RMB 10,000**, 2019.
- China College Students Service Outsourcing Innovation and Entrepreneurship Competition}, funded with **RMB 8,000**, 2018.

TECHNICAL SKILLS

Programming: Python, C, R, SQL, Linux, Git, Docker, FastAPI, React, Spring Boot, Latex.

Deep Learning Frameworks: PyTorch, TensorFlow / Keras, scikit-learn.

AI models: CNN, 3D CNN, U-Net, Transformer, BiLSTM, KANs, XGBoost, Prophet, LSTM, GRU.

LANGUAGE SKILLS

English: IELTS 6.5; experienced in English-medium study, academic reading, technical writing, and cross-cultural collaboration.

Mandarin: Native.